Developing a Low Cost Plasma Thruster

Olin Plasma Engineering & Electric Propulsion Team

Braden Oh, Albert Countryman, Avery Clowes, Lily Dao, James Jagielski, and Ben Eisenbraun

Overview

- Who We Are
- Introduction to Space Propulsion
- How Hall Thrusters Work
- Our Design
- Achieving Ignition
- The Need to Increase Access
- Conclusion

- Independent Study (ISR) Team
  - **Project based, self-directed** learning enabled by Olin College
  - **No funding** awarded

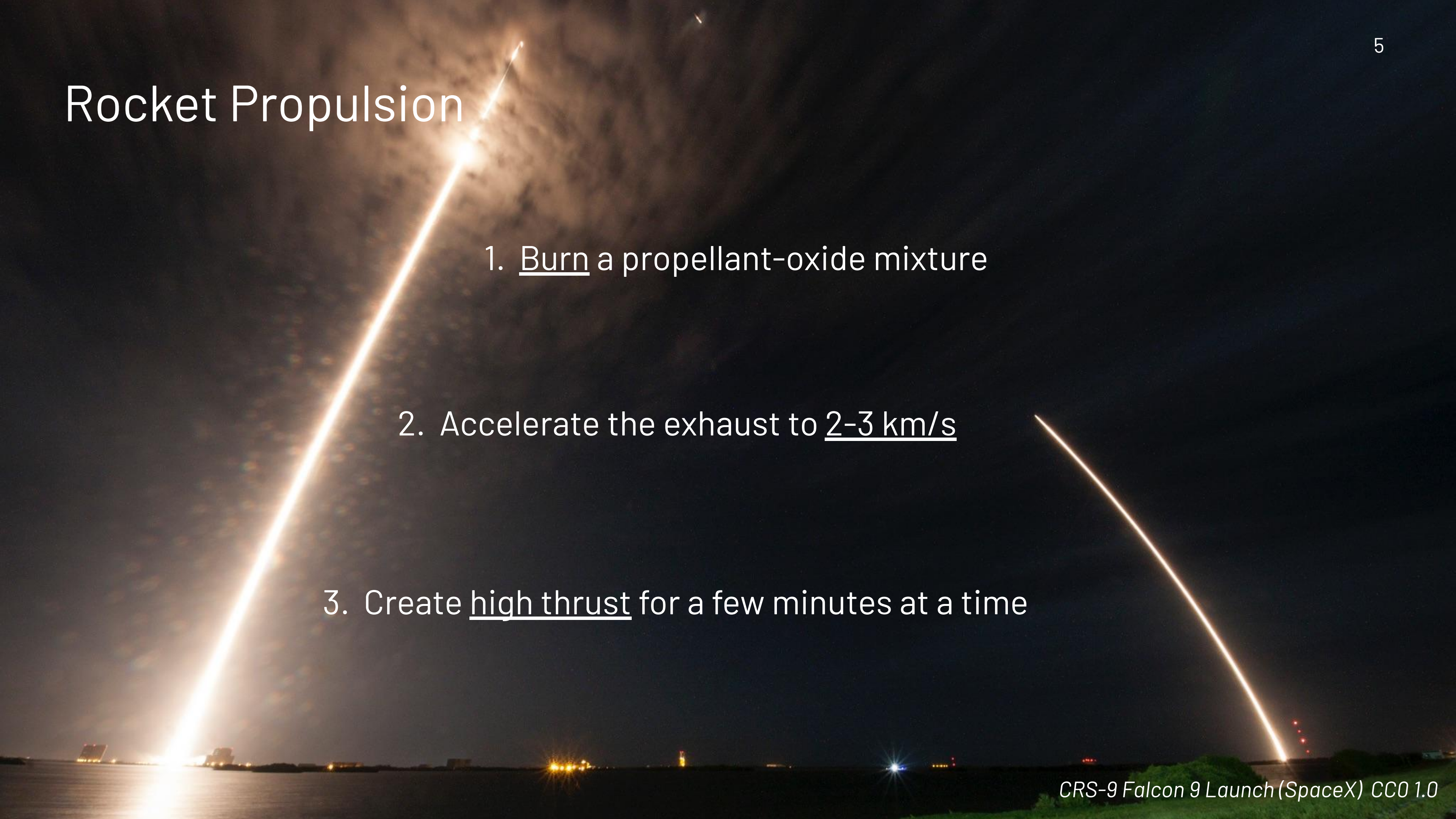
- **7 undergrads**, three schools

  + Prof. Chris Lee, ISR sponsor

- We want to enable space exploration!
  - Engineer plasma propulsion devices
  - **Create resources** for others

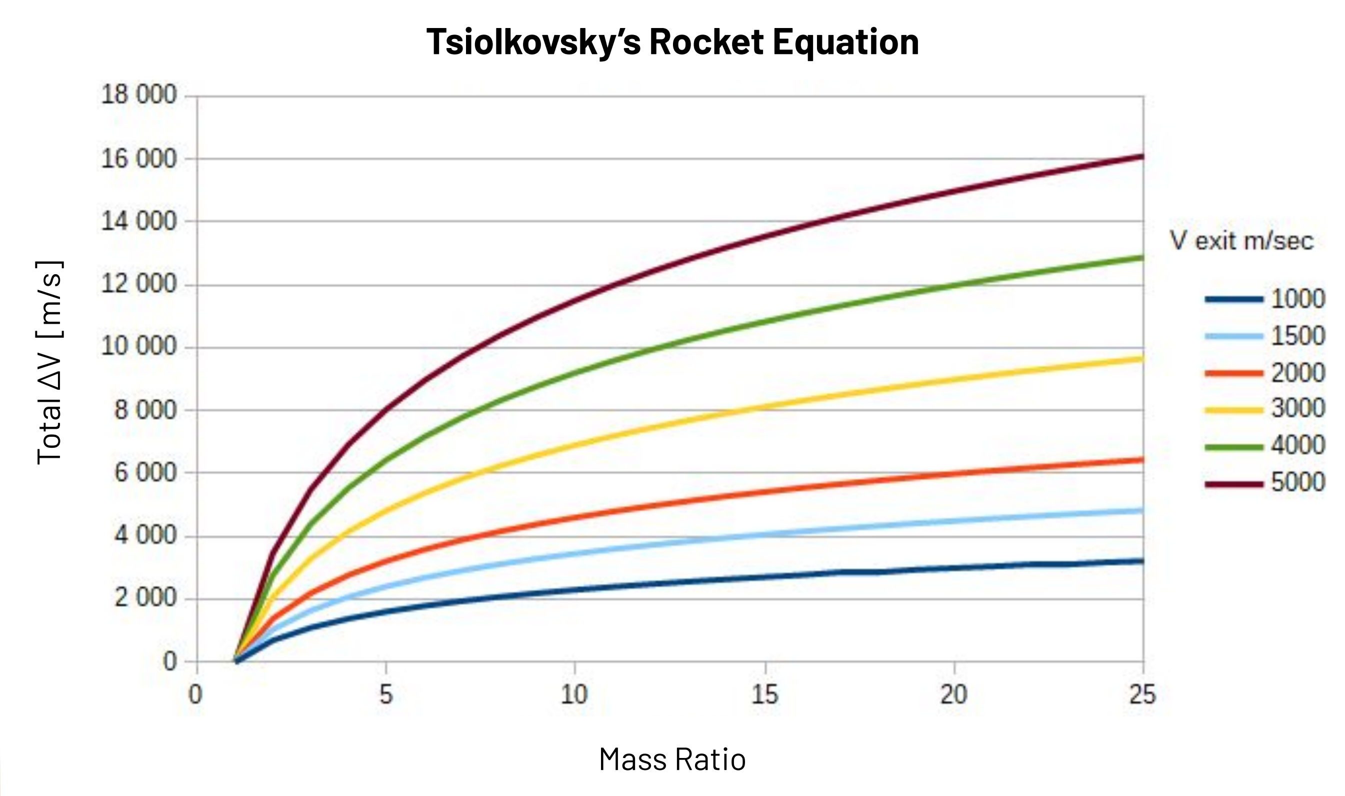
Introduction to Space Propulsion

Rocket Propulsion



1. Burn a propellant-oxide mixture
2. Accelerate the exhaust to 2-3 km/s
3. Create high thrust for a few minutes at a time

CRS-9 Falcon 9 Launch (SpaceX) CC0 1.0

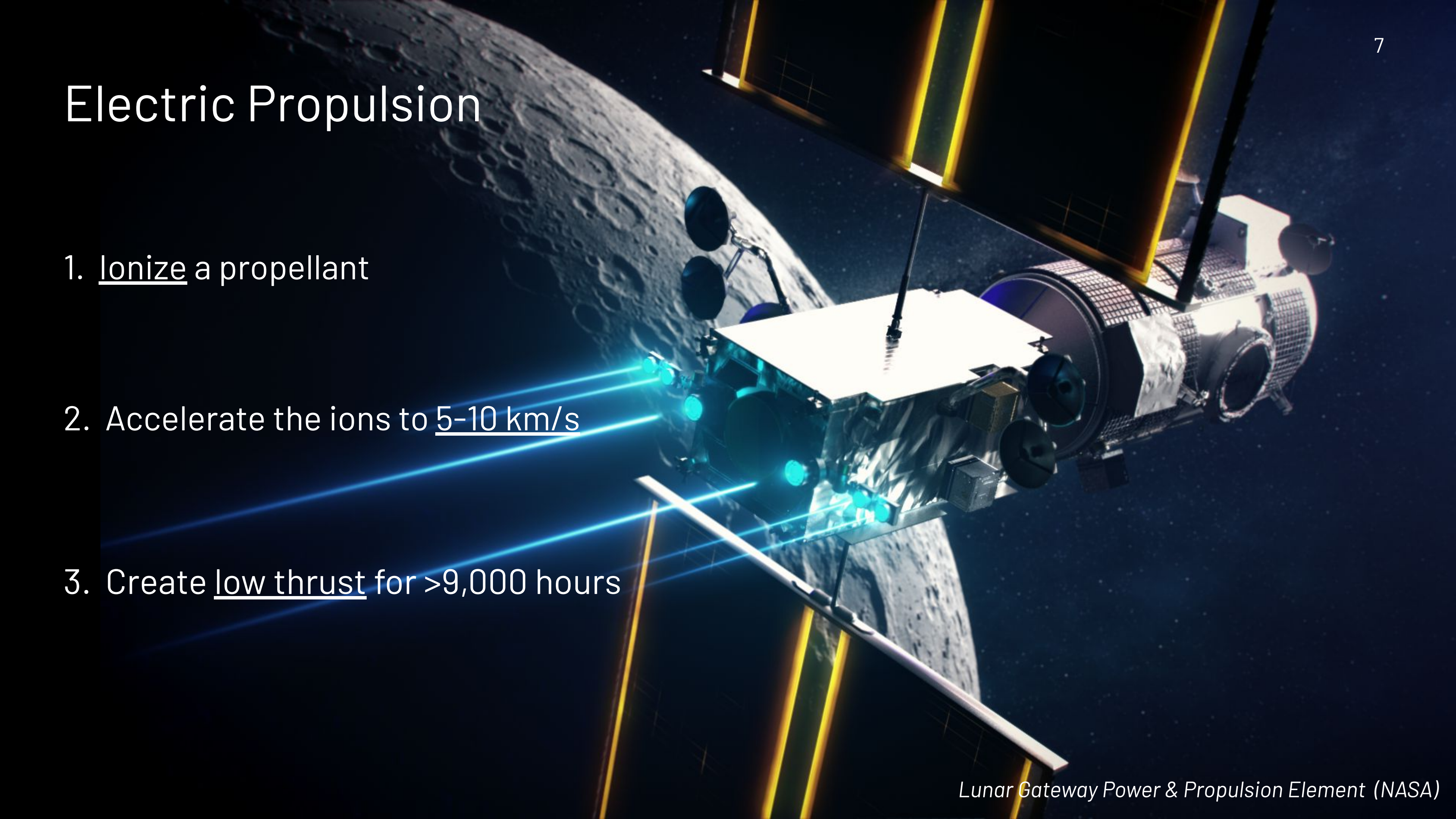

Tsiolkovsky's Rocket Equation
18 000
16 000
14 000
12 000
10 000
8 000
6 000
4 000
2 000
0
Total ΔV [m/s]
0
5
10
15
20
25
Mass Ratio
V exit m/sec
1000
1500
2000
3000
4000
5000

Electric Propulsion

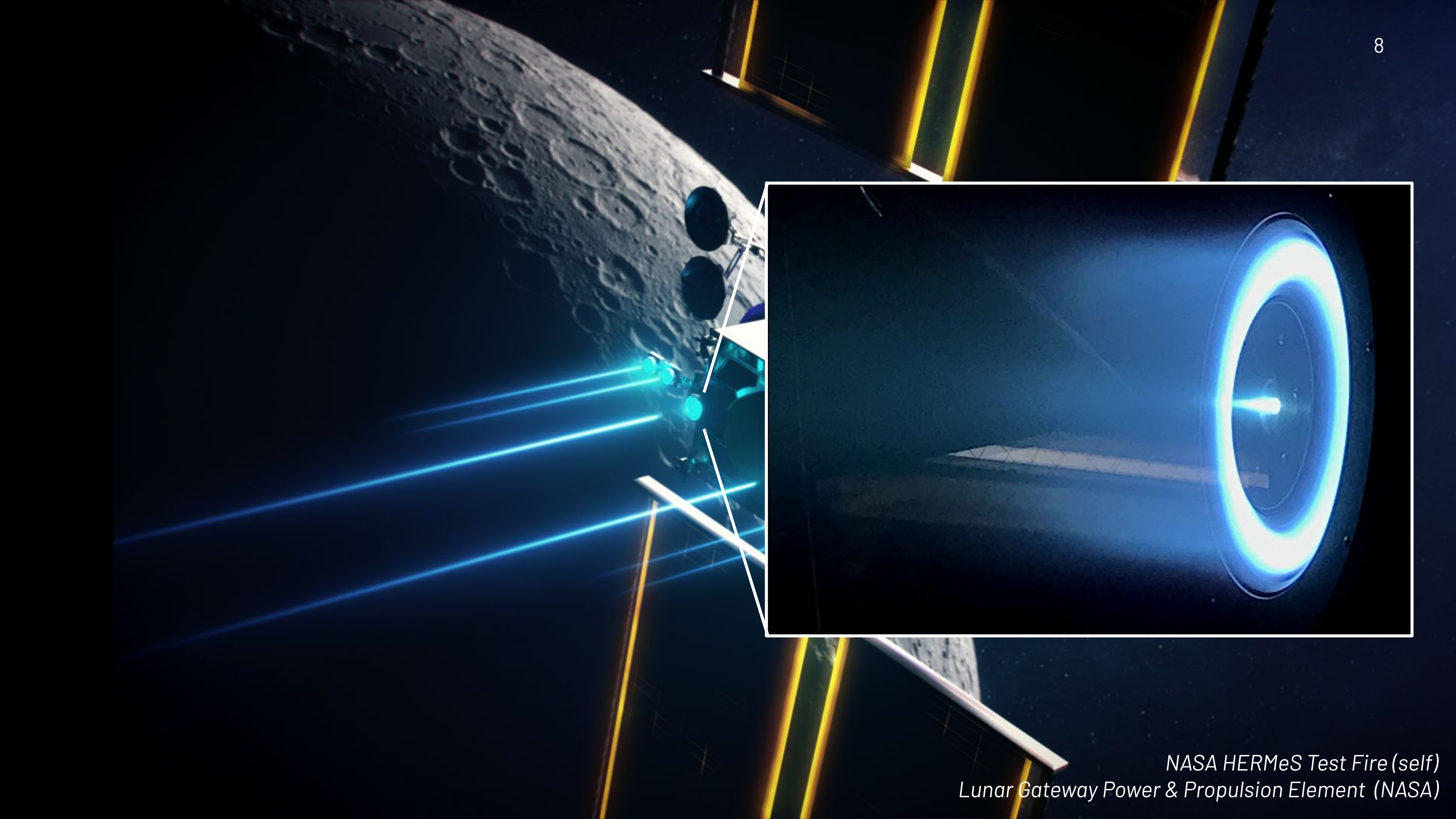
1. <u>Ionize</u> a propellant

2. Accelerate the ions to <u>5-10 km/s</u>

3. Create <u>low thrust</u> for >9,000 hours

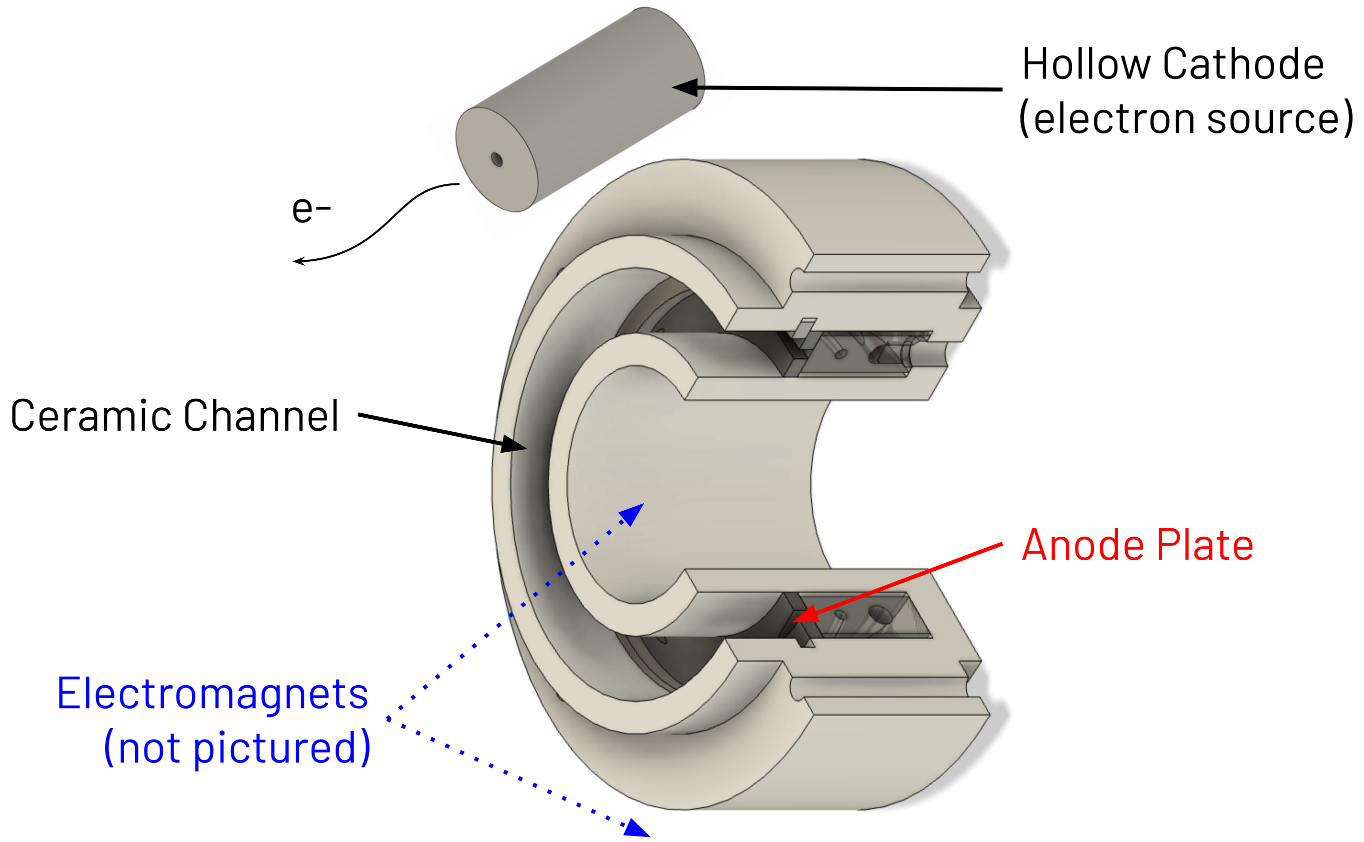


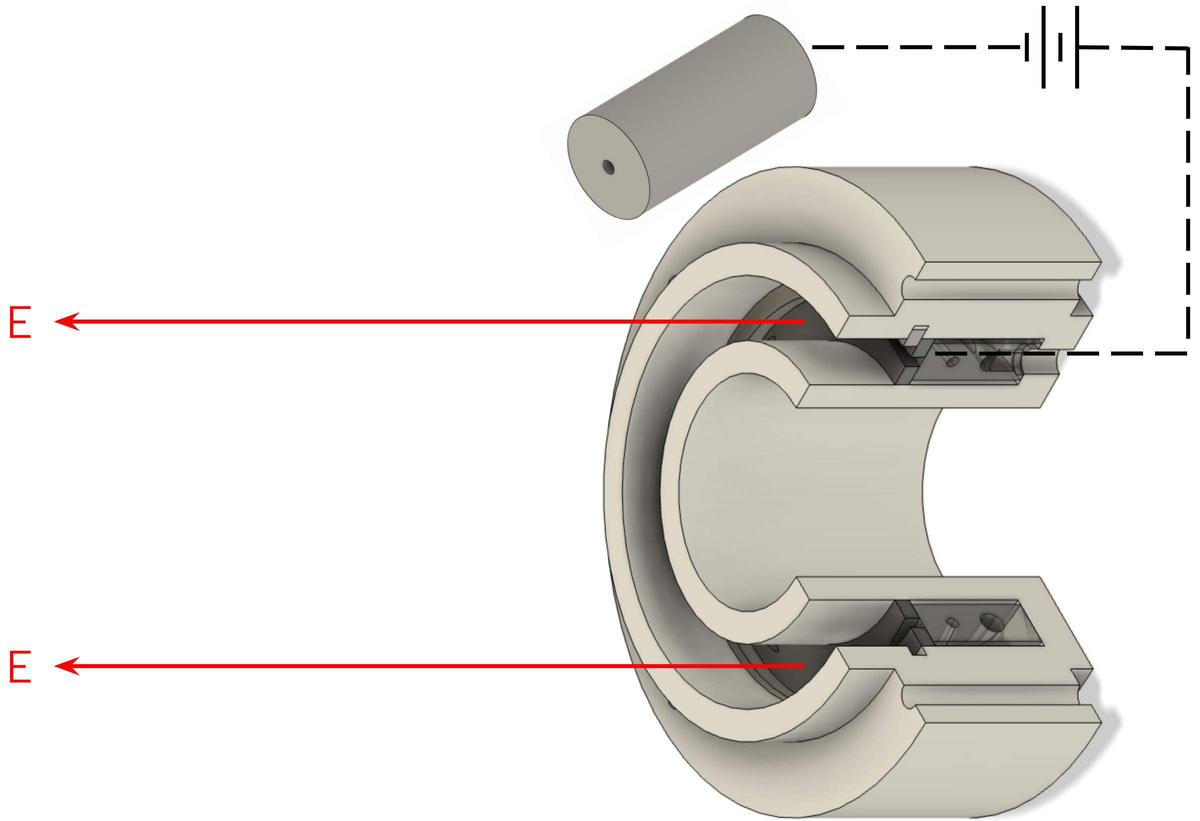
Lunar Gateway Power & Propulsion Element (NASA)



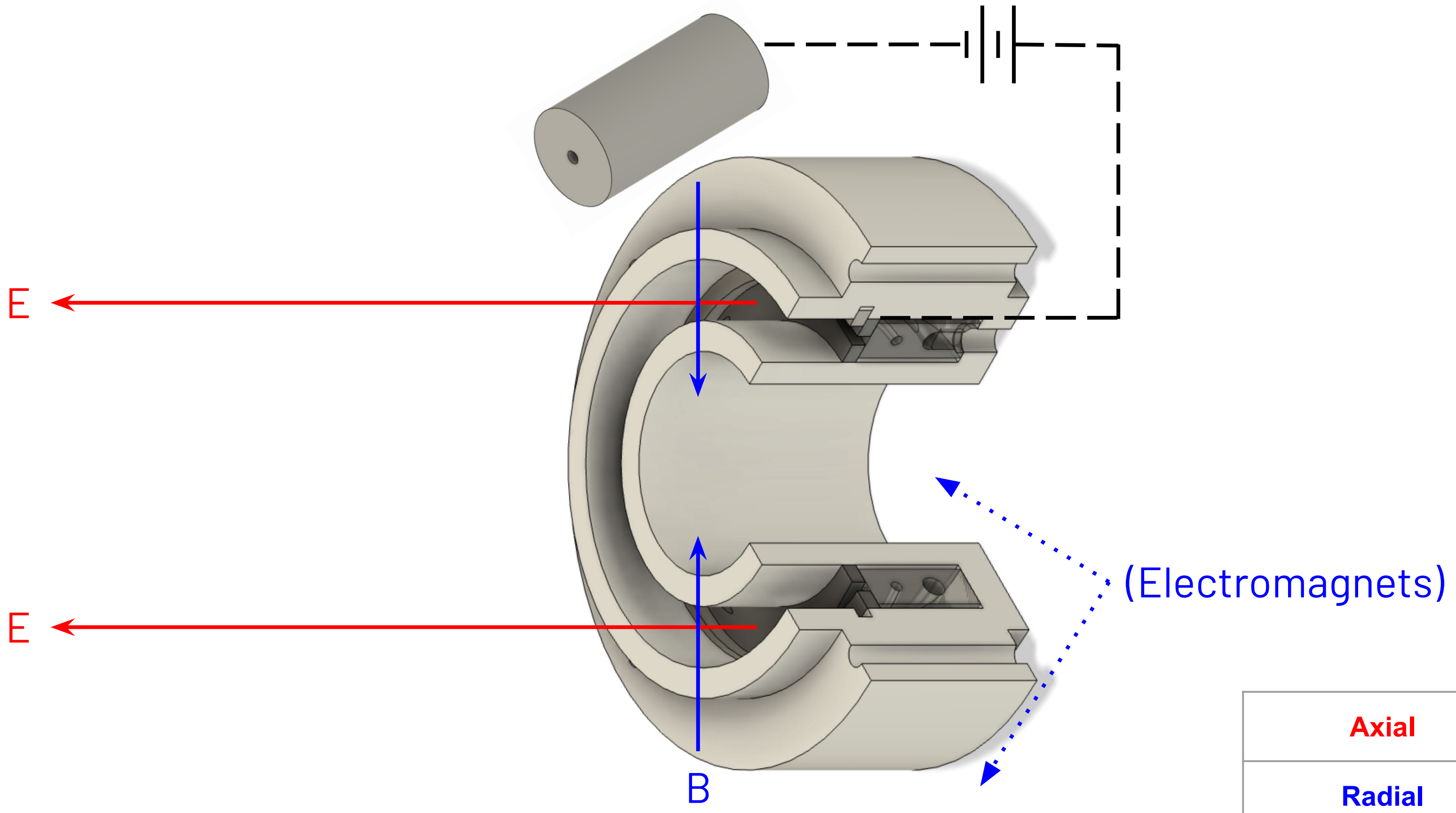
NASA HERMeS Test Fire (self)
Lunar Gateway Power & Propulsion Element (NASA)

How Hall Thrusters Work

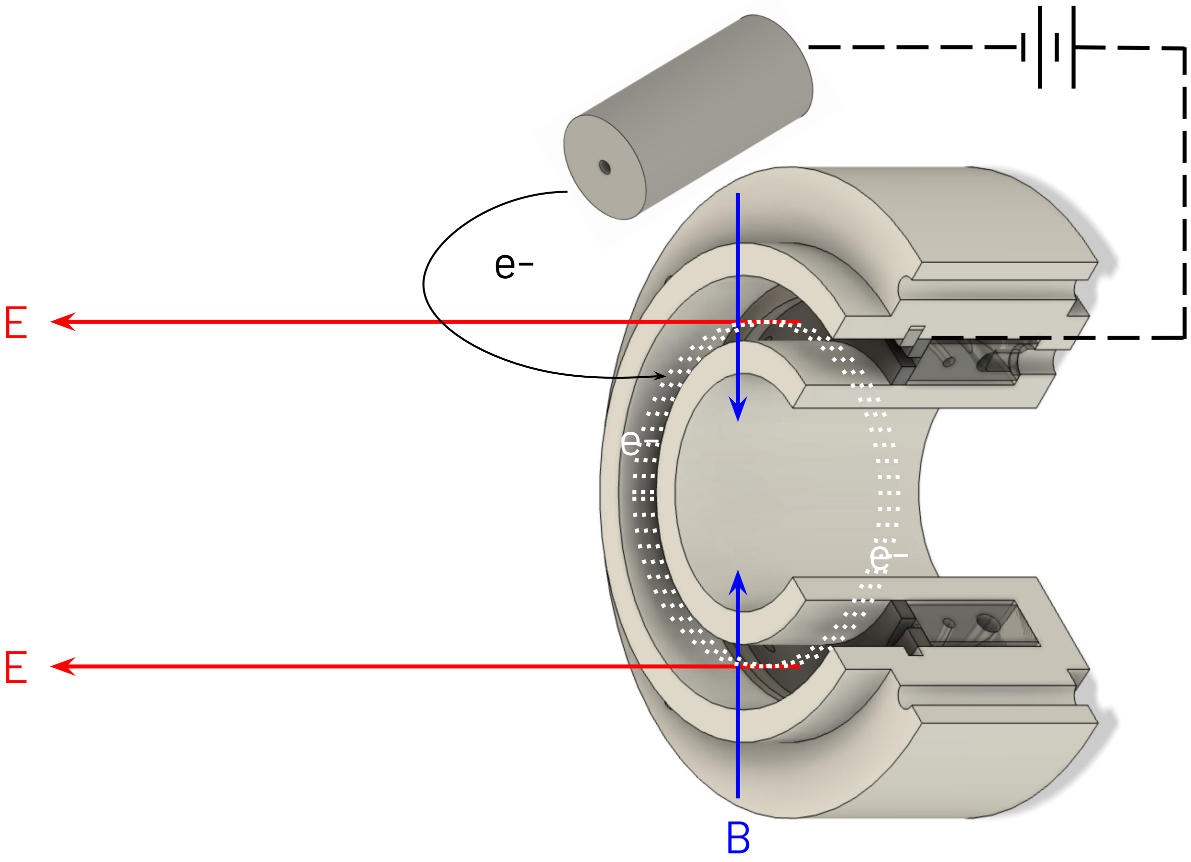

Hollow Cathode
(electron source)
e-
Ceramic Channel
Anode Plate
Electromagnets
(not pictured)



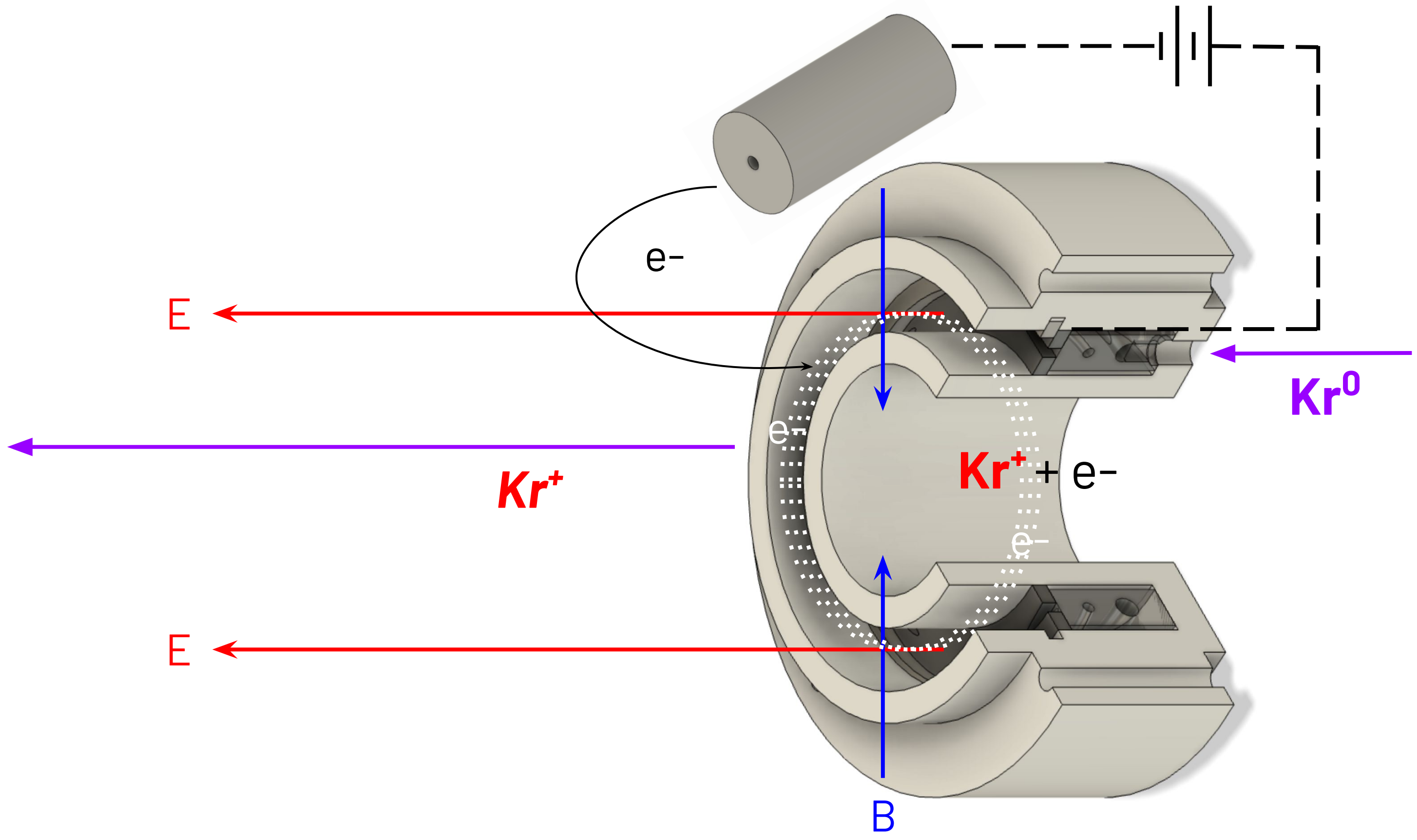
Axial



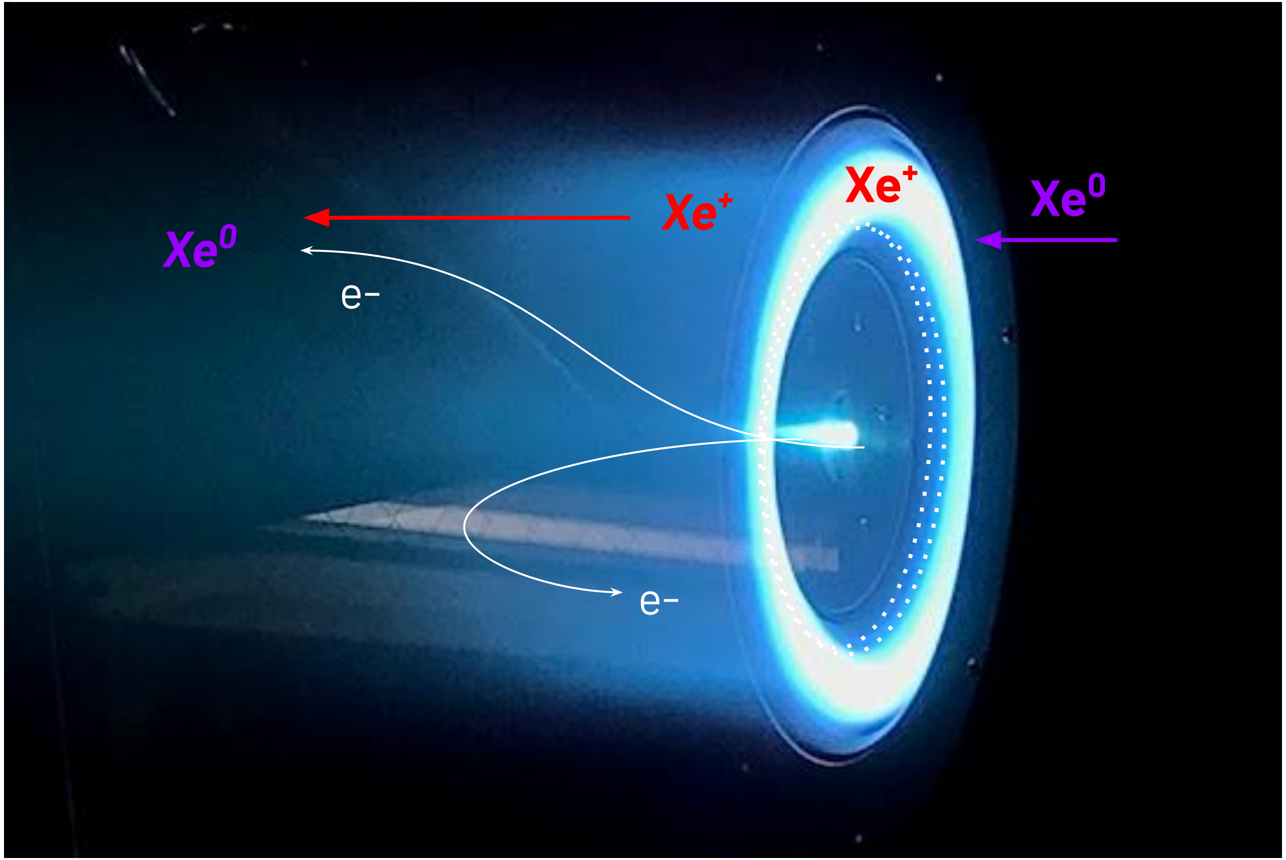
Axial
Radial



Axial
Radial
Azimuthal

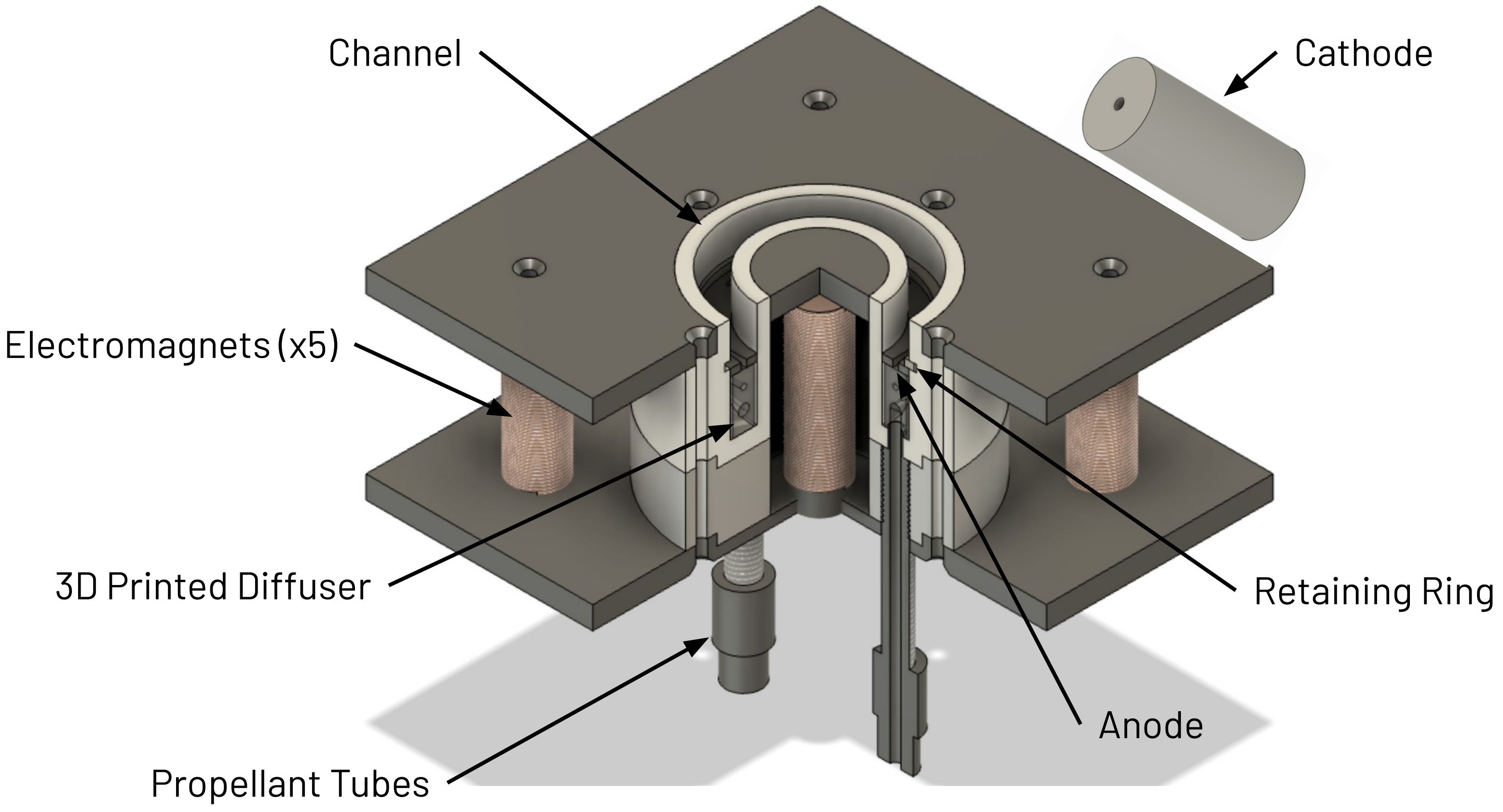


Axial
Radial
Azimuthal

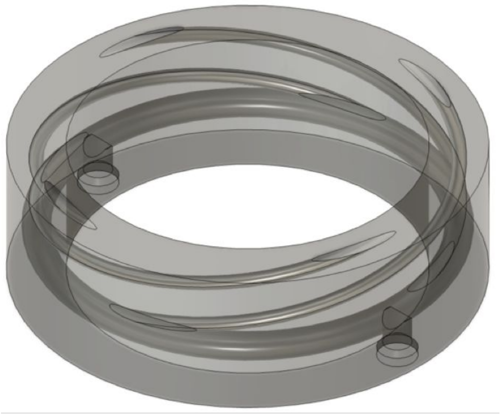


NASA HERMeS Test Fire (self)

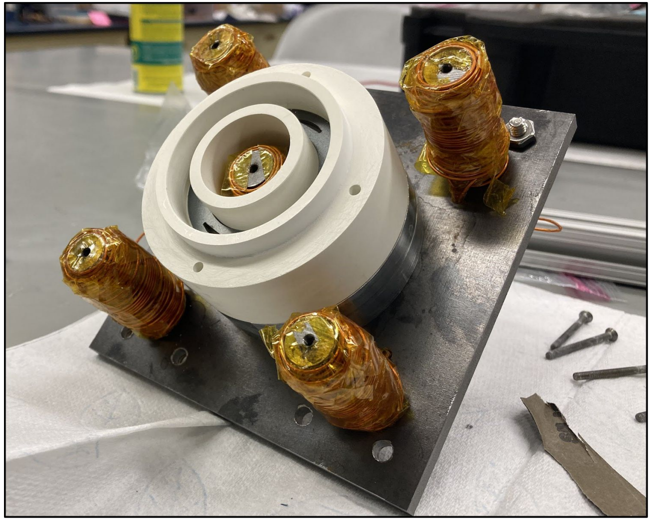
Our Design



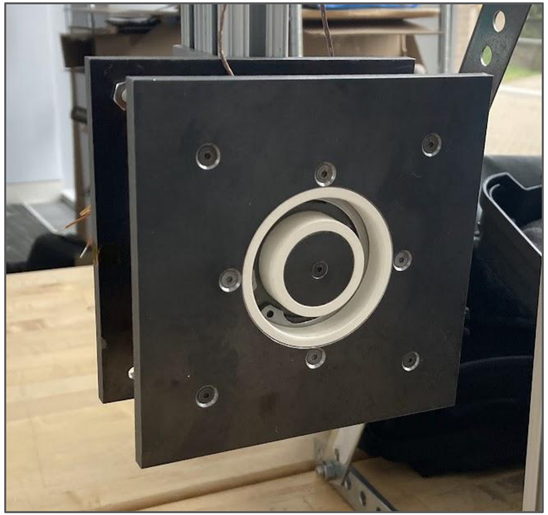
Channel
Cathode
Electromagnets (x5)
3D Printed Diffuser
Retaining Ring
Anode
Propellant Tubes



3D printed gas diffuser*

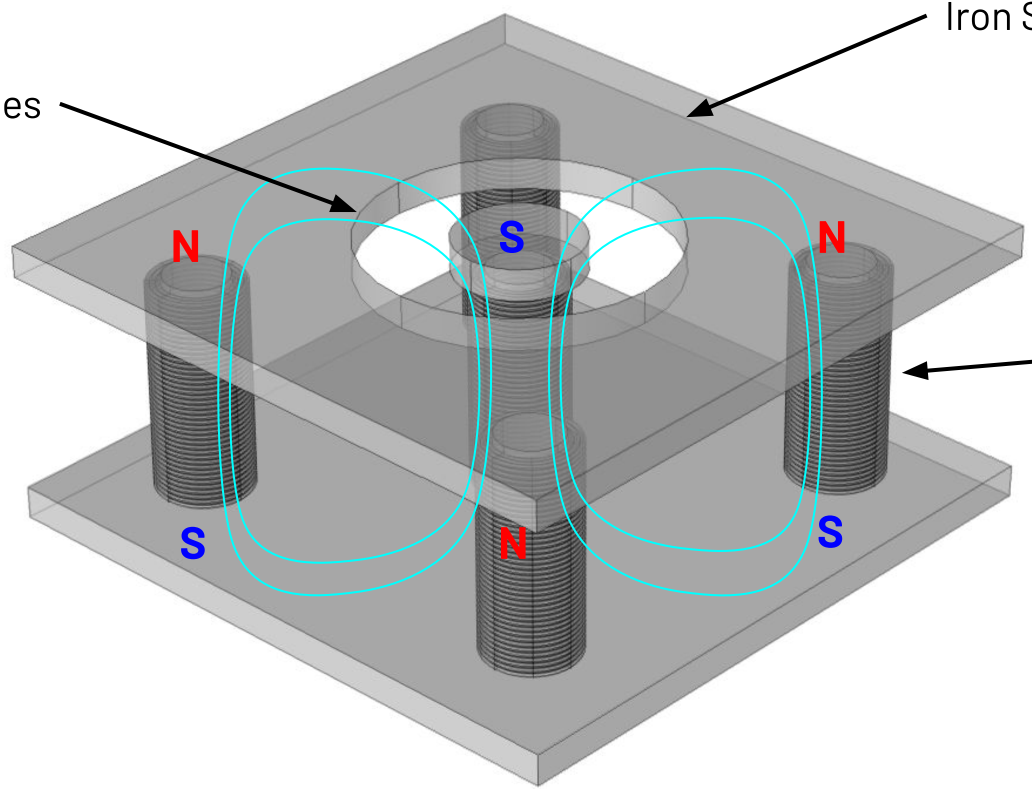


Partially assembled thruster



Assembled on thrust stand

* Oh et al. *Journal of Electric Propulsion* (2023) 2:6



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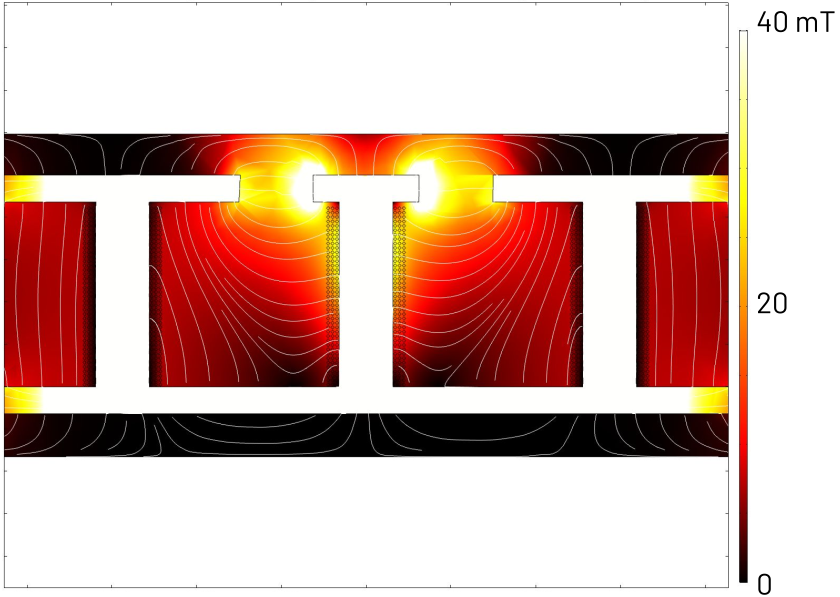
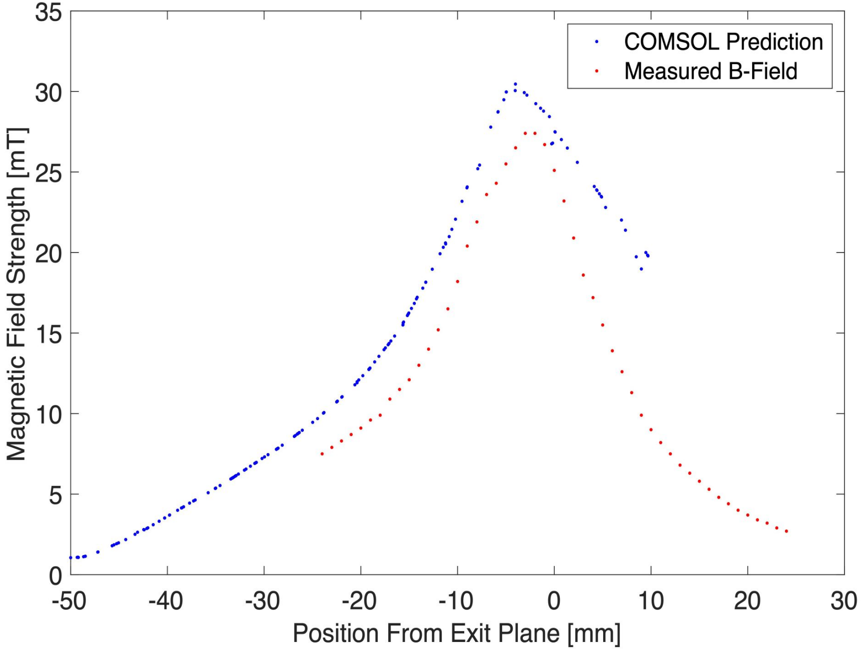
COMSOL Magnetic Field Simulation



Predicted vs. Actual Field Strength



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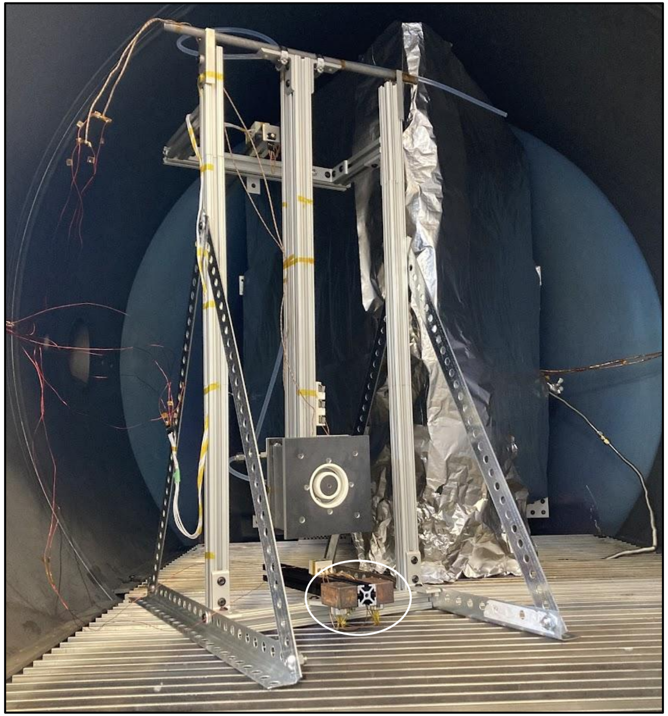
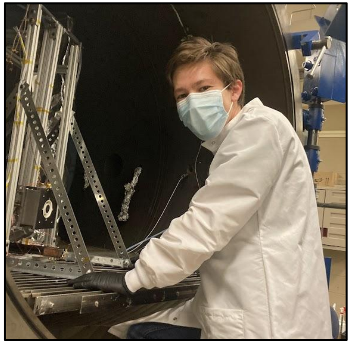
Ignition Attempt (May 2022)



Thruster at MIT Space Propulsion Lab*



Action photos!

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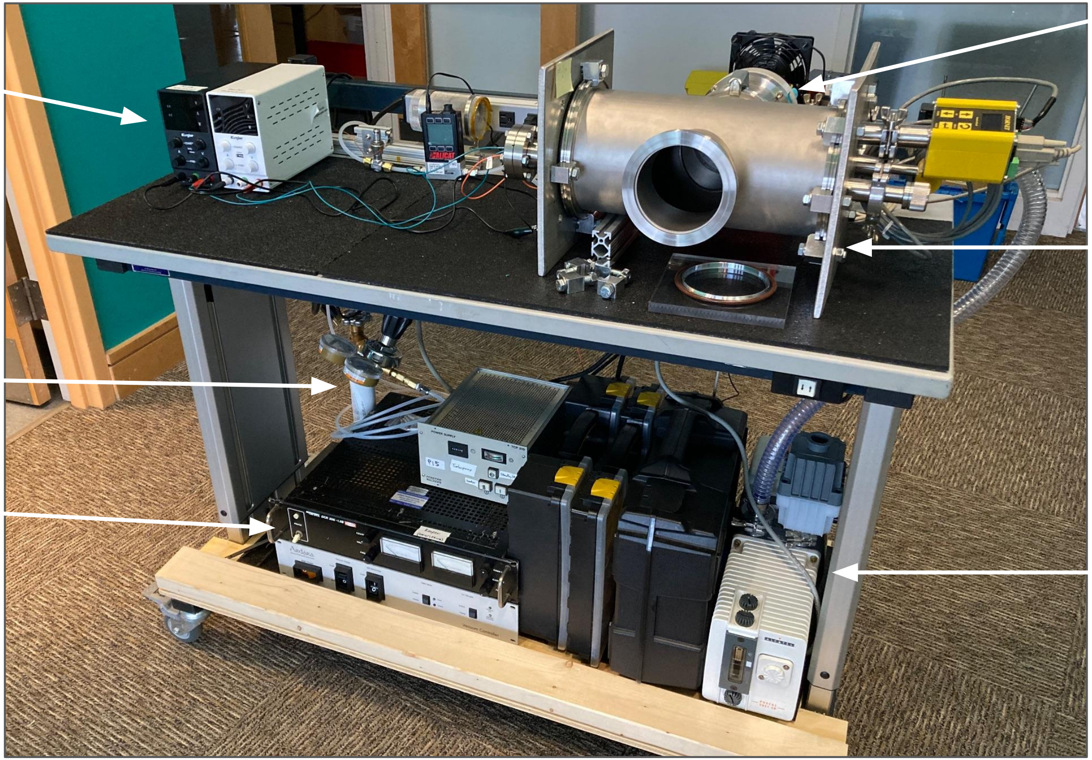
High-Vacuum Infrastructure

5A / 10A supplies

Kr / Ar tank

300V supply

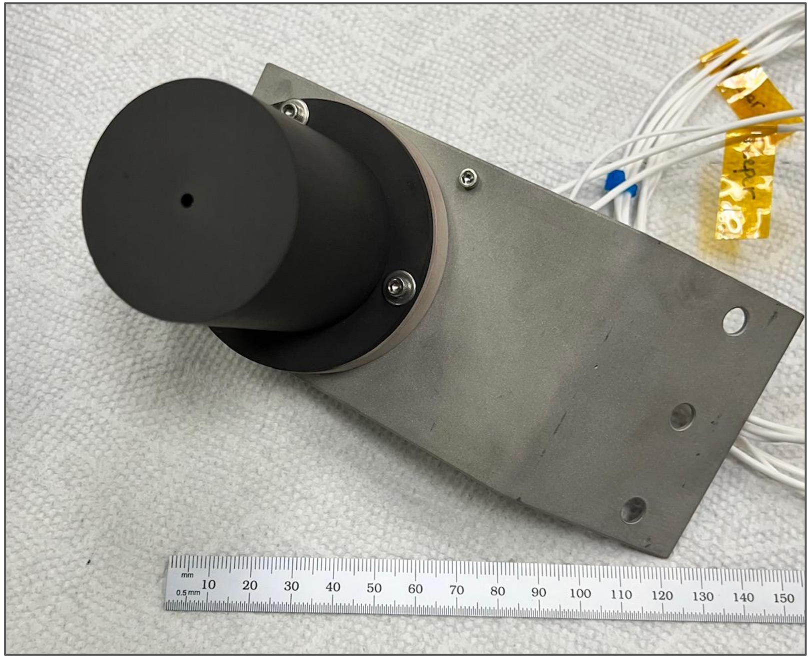


Turbopump

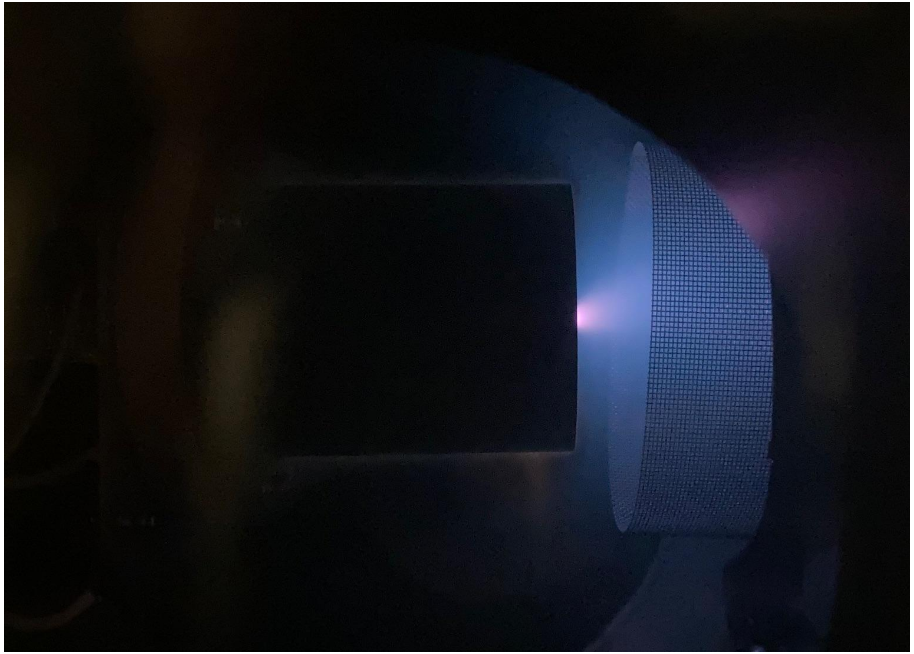
6″ diameter chamber

Dual-vane
roughing pump

Hollow Cathode Testing



Cathode donated by Plasma Controls LLC

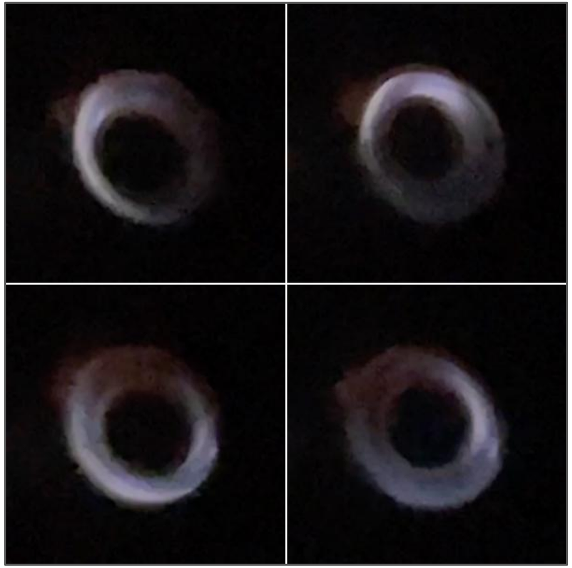


Argon plasma test in Olin chamber

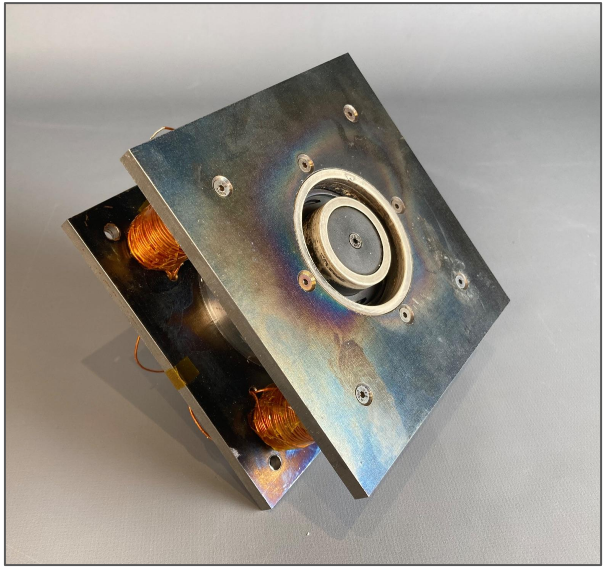
Ignition! (Feb 2023)

Peak power: 200W

~ 2 hrs continuous operation



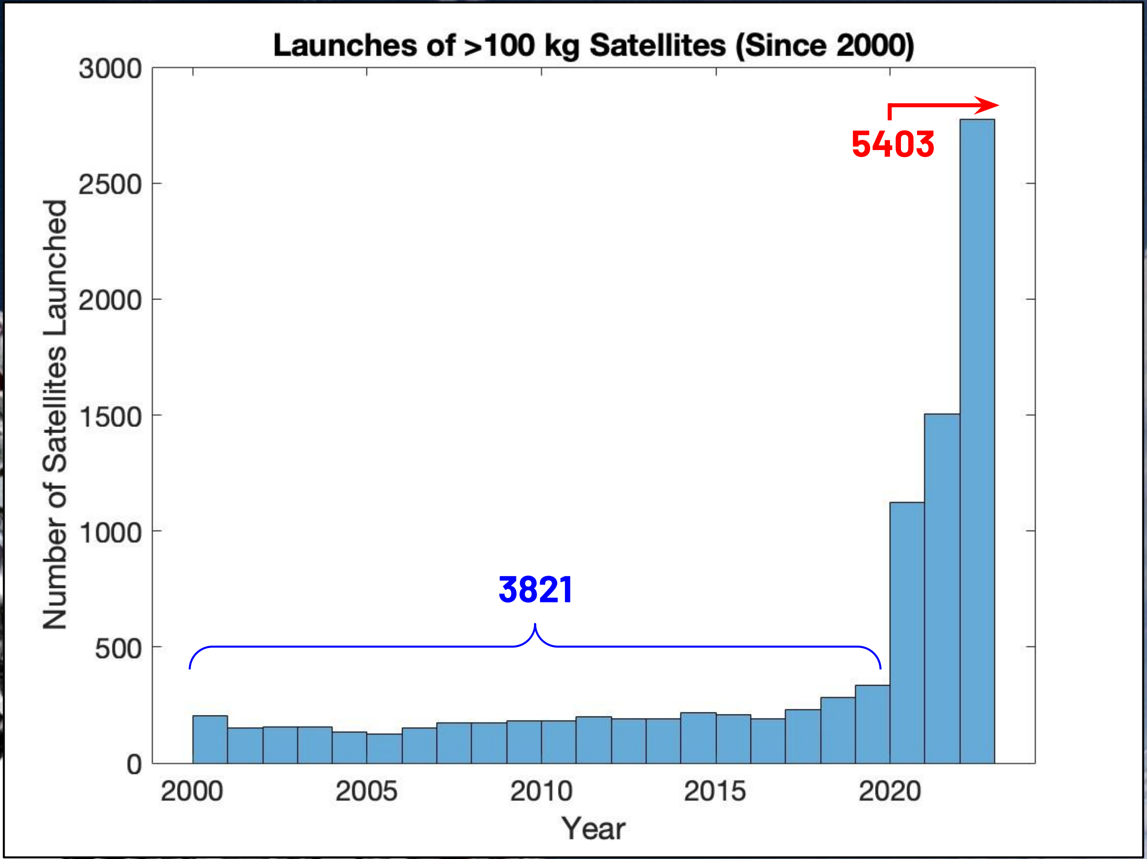
Characteristic rotating spoke instability



Thermal discoloration

Asymmetric diffuser heating

The Need to Increase Access



Graph data from Prof. Jonathan McDowell (Harvard)

Stack of 60 Starlink Satellites (SpaceX) CC0 1.0

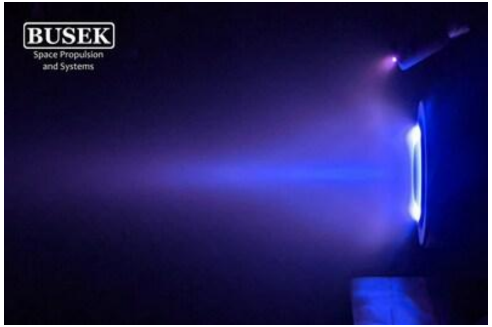
The Hall Thruster Landscape

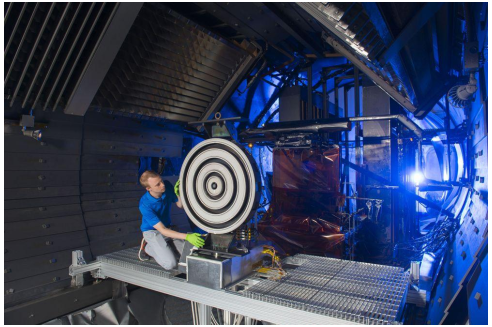
Busek Co.

Astra / Apollo Fusion

~~SpaceX Starlink~~

~~ROSKOSMOS Fakel~~





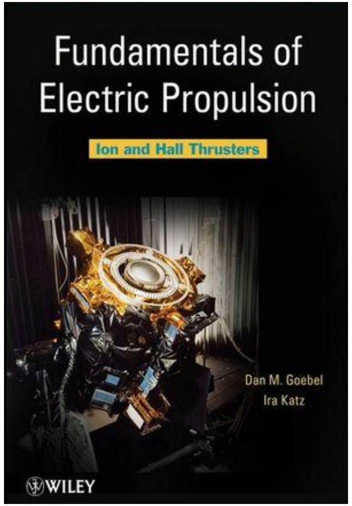
University of Michigan

Georgia Tech

NASA JPL **NASA Glenn**

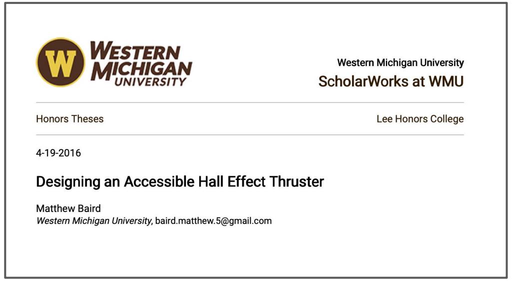
Image Credit:
Busek Co (website), NASA, JPL-Caltech

Accessible Literature



Graduate physics textbook
(Wiley, 2008)

?



Western Michigan University
ScholarWorks at WMU

Honors Theses | Lee Honors College

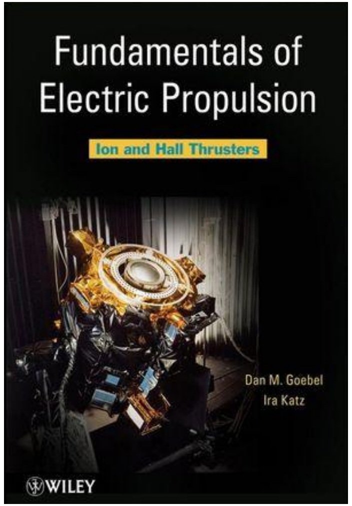
4-19-2016

Designing an Accessible Hall Effect Thruster

Matthew Baird
Western Michigan University, baird.matthew.5@gmail.com

Undergraduate design thesis
(WMU, 2016)

Accessible Literature



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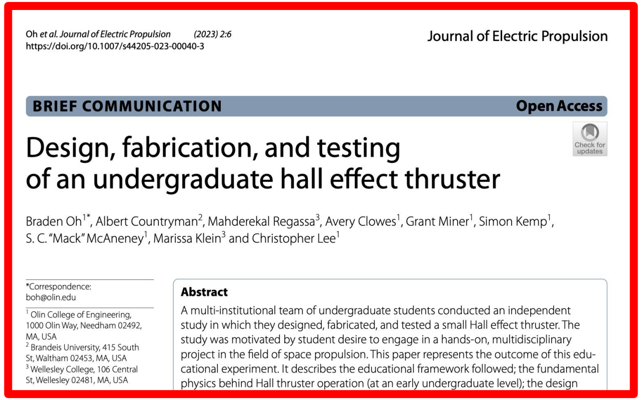


Oh *et al. Journal of Electric Propulsion* (2023) 2:6
https://doi.org/10.1007/s44205-023-00040-3

Journal of Electric Propulsion

BRIEF COMMUNICATION Open Access

Design, fabrication, and testing of an undergraduate hall effect thruster

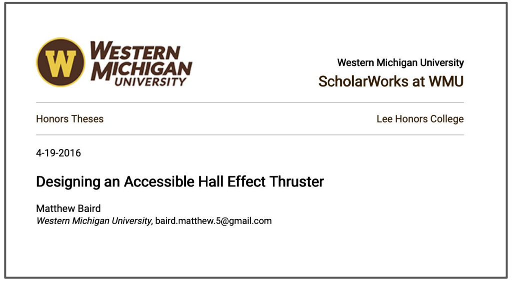
Braden Oh[1*], Albert Countryman[2], Mahderekal Regassa[3], Avery Clowes[1], Grant Miner[1], Simon Kemp[1], S. C. "Mack" McAneney[1], Marissa Klein[3] and Christopher Lee[1]

*Correspondence: boh@olin.edu
[1] Olin College of Engineering, 1000 Olin Way, Needham 02492, MA, USA
[2] Brandeis University, 415 South St, Waltham 02453, MA, USA
[3] Wellesley College, 106 Central St, Wellesley 02481, MA, USA

Abstract
A multi-institutional team of undergraduate students conducted an independent study in which they designed, fabricated, and tested a small Hall effect thruster. The study was motivated by student desire to engage in a hands-on, multidisciplinary project in the field of space propulsion. This paper represents the outcome of this educational experiment. It describes the educational framework followed; the fundamental physics behind Hall thruster operation (at an early undergraduate level); the design

Engineering design walkthrough
(JEP, 2023)

→



Western Michigan University

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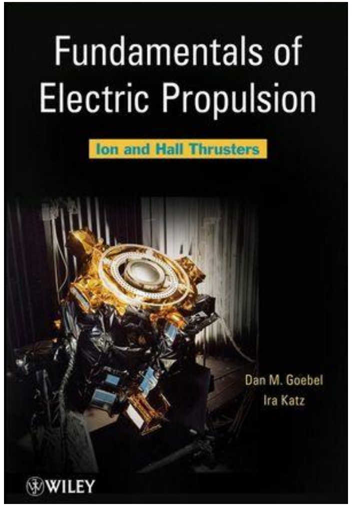

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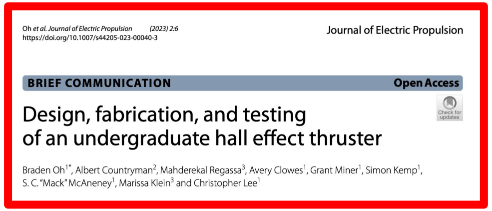
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Journal of Electric Propulsion

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Engineering design walkthrough
(JEP, 2023)


127th ASEE Annual Conference

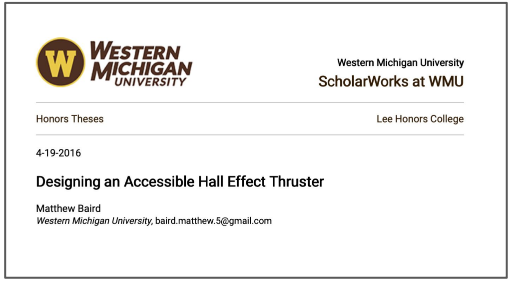
June 21-24, 2020 | Palais des congrès de Montréal | Montréal, Canada

Paper ID #29672

(Student Paper) Undergraduate Demonstration of a Hall Effect Thruster: Self Directed Learning in an Advanced Project Context

Braden K. Oh, Olin College of Engineering
Justin Haruaki Kunimune, Olin College of Engineering
Jonah Spicher, Olin College of Engineering
Lauren Anfenson, Franklin W. Olin College of Engineering
Dr. Rebecca Christianson, Draper Labs

Undergraduate curriculum
(ASEE, 2020)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Undergraduate design thesis
(WMU, 2016)

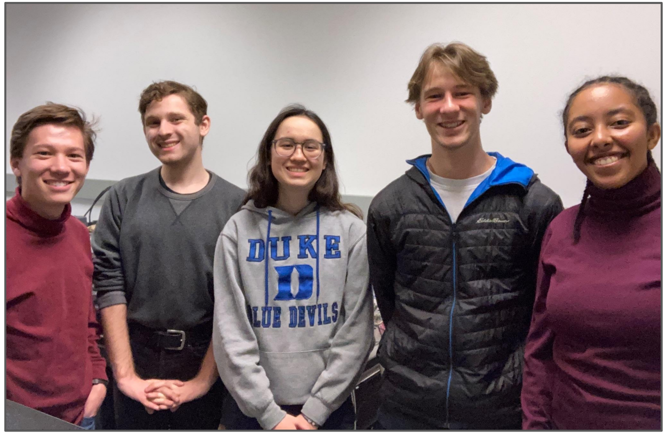
Conclusions

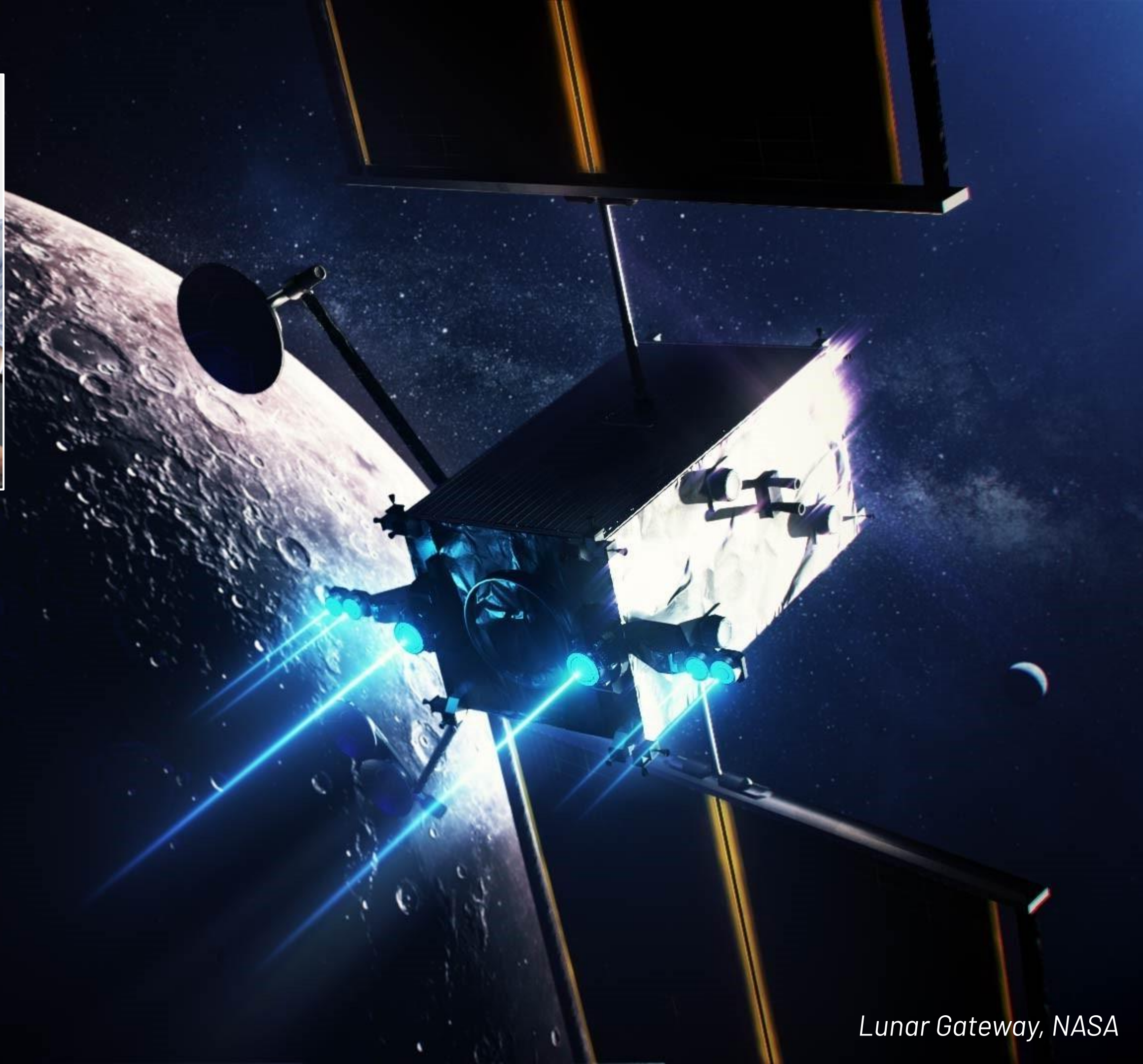
- **Industry needs more** engineers experienced in electric propulsion.

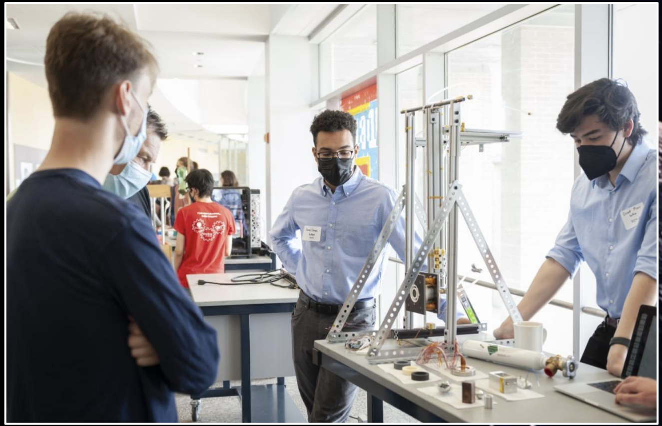
- Electric propulsion **is accessible,** even in undergraduate curricula.

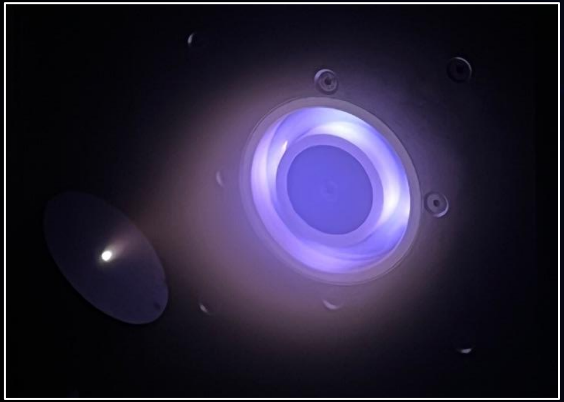
- Accessible **literature is immediately** usable by fledgling teams.

- **We can, right now,** be a part of building the road to space!









Lunar Gateway, NASA

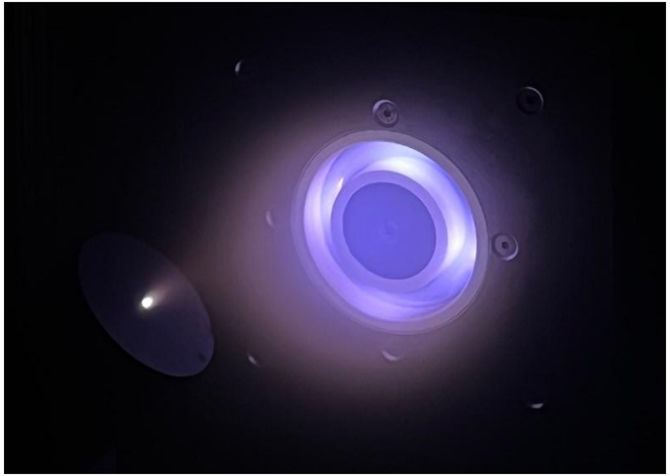
Contingency

We are the first, but we won’t be the last!

- Electric propulsion **is accessible** to undergraduate students.
- **Hands-on** EP projects can fit into engineering curricula.
- Our papers are **already being utilized** by fledgling teams.

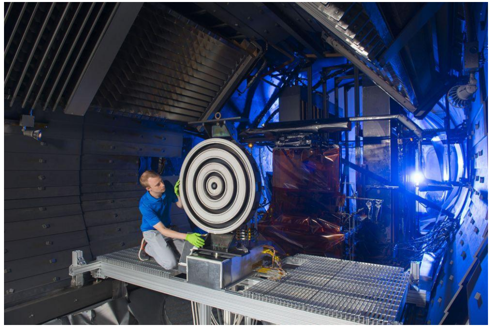
Ongoing Work

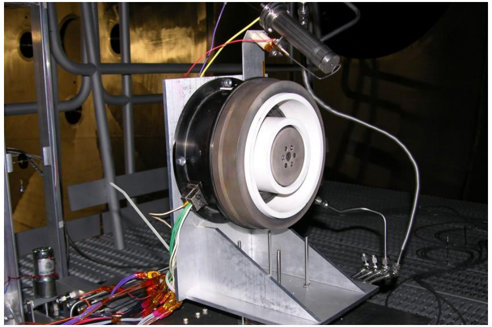
- 3D printed parts
  - Diffuser thermal management
  - Entire thruster channels
- Low-cost cathodes
  - Heaterless cathode experiments
- Magnetic shielding
  - Next-gen, long life Hall thruster
- Continue creating accessible resources



Graduate-Level Research (HET)

University of Michigan*
(credit NASA)





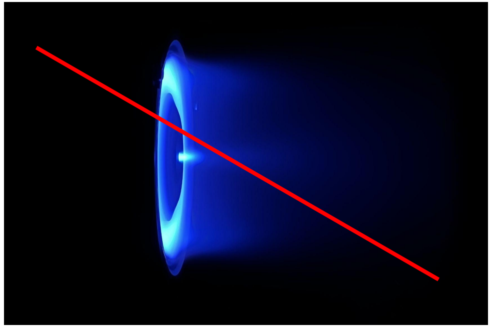
Georgia Tech
(credit HPEPL website)

NASA JPL*
(credit NASA/JPL-Caltech)

* Where new thruster design presently occurs

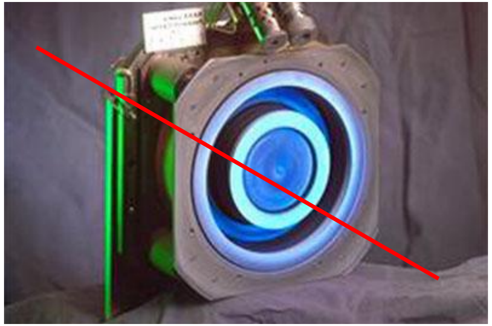
Private Industry (HET)

Busek Co.
(credit Busek website)



SpaceX Starlink
(credit SpaceX, via LinkedIn)

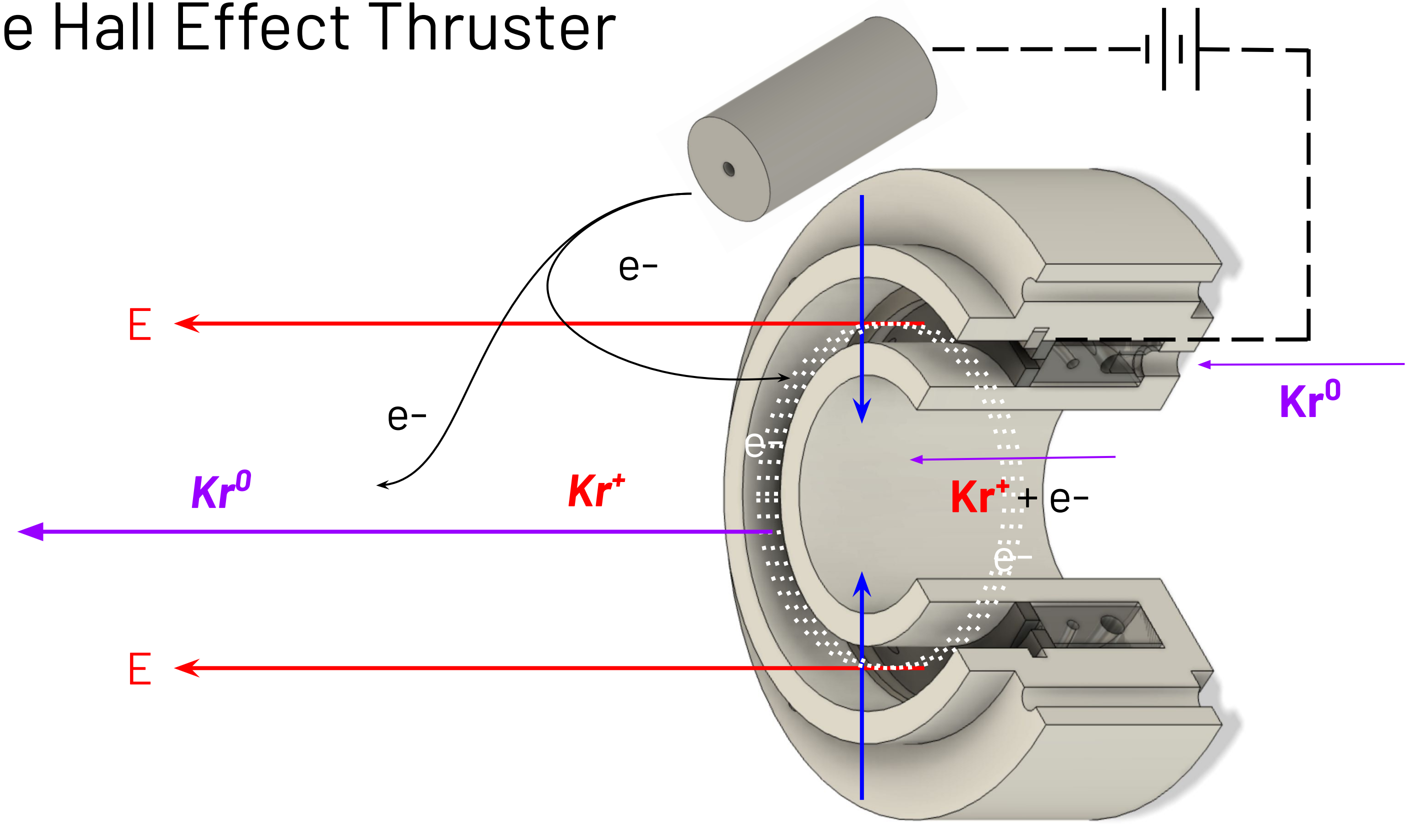
Astra / Apollo Fusion
(credit NASA/JPL-Caltech)



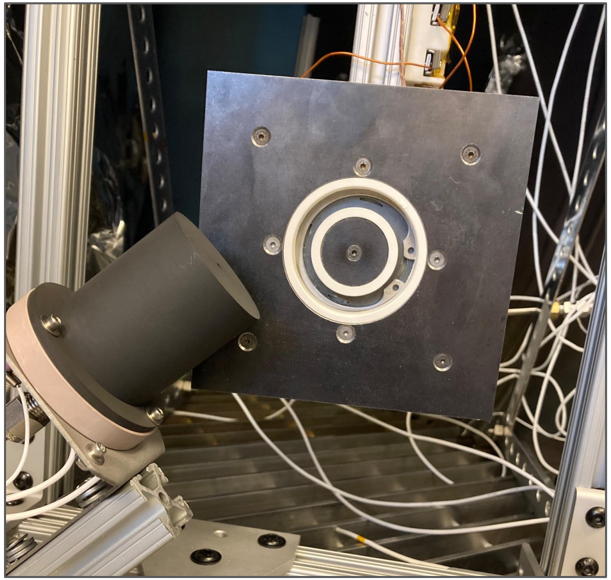
ROSKOSMOS Fakel
(credit U-M PEPL)

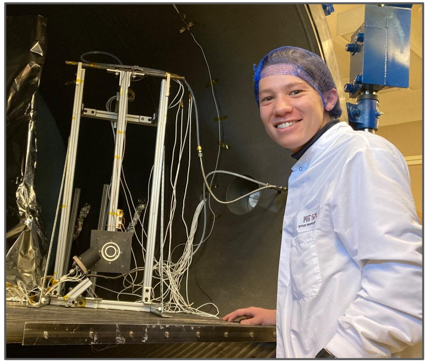
\ Not for sale to U.S. Customers

The Hall Effect Thruster



Returning to MIT

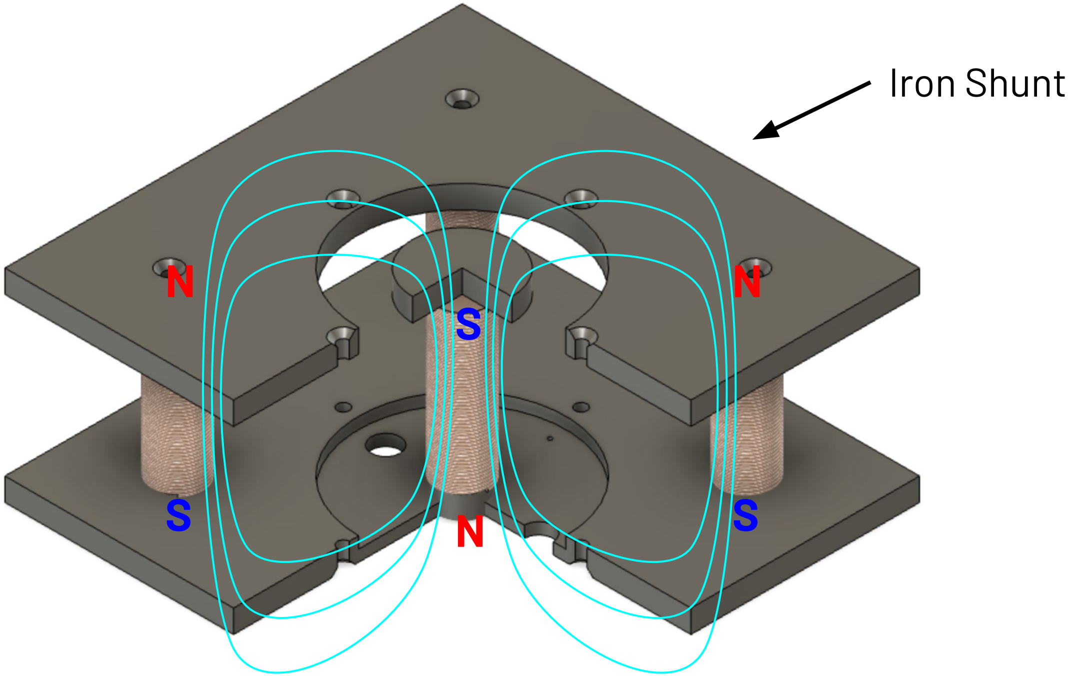




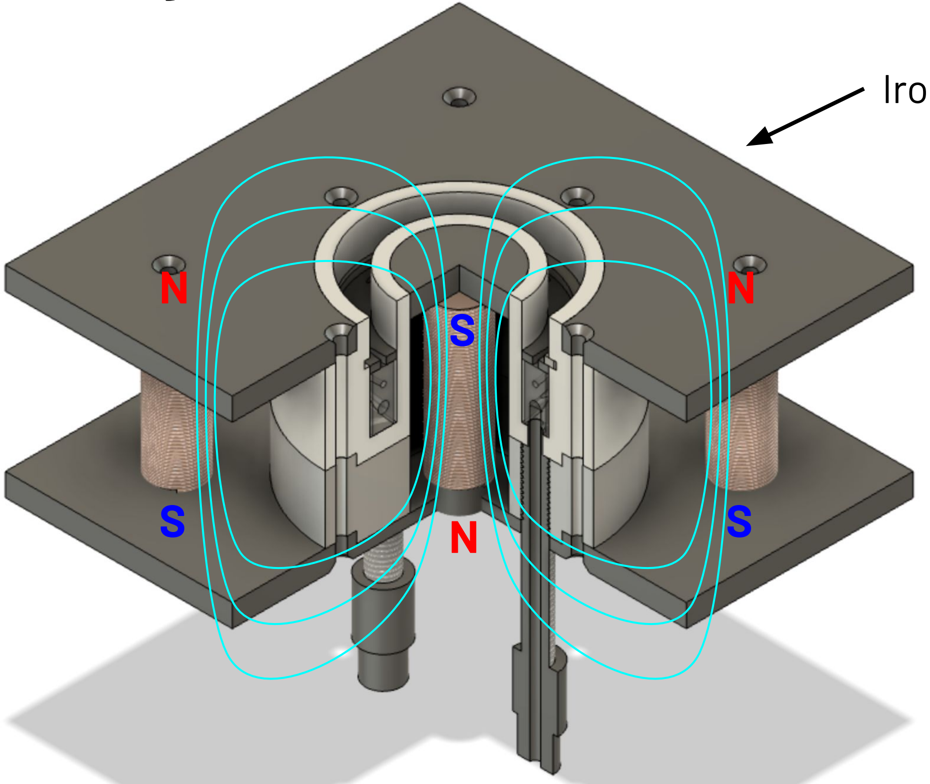
Meet the Team

- 8 self-directed, undergraduate students
  - 5 engineers + 3 physicists
  - 4 sophomores + 3 juniors + 1 senior
- Spanning three colleges
  - 5 from Olin College
  - 2 from Wellesley College
  - 1 from Brandeis University
- Technical and educational goals
  - Engage in the self-directed research process
  - Create an entry-level resource for undergrads

Magnetic Field Design

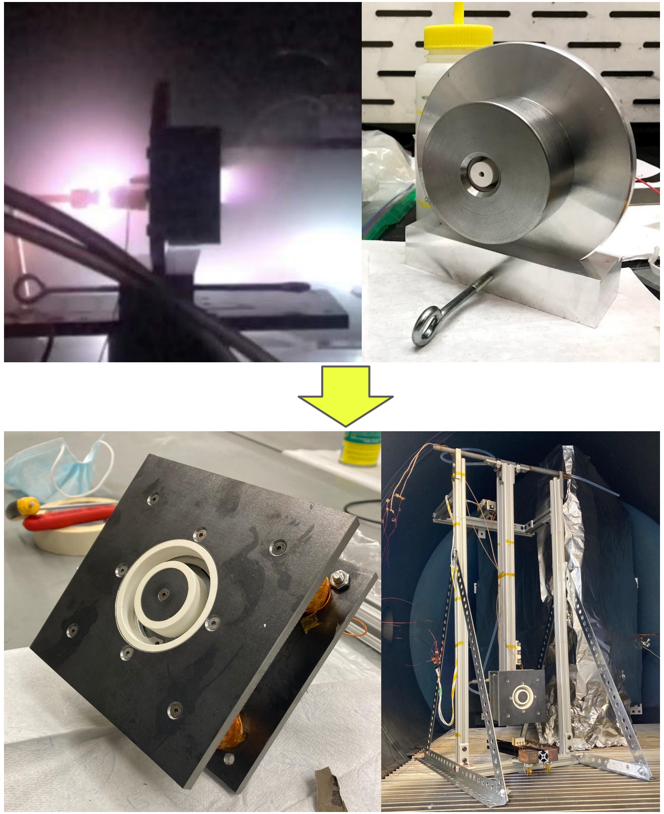


Magnetic Field Design



Project Objectives

1. Create hands-on, multidisciplinary course to engage in the research process
2. Improve upon prior Olin College HET
3. Develop entry-level HET resource



Outcome: Failure to achieve ignition, but wrote an entry-level resource and conducted common research activities